

Data Warehousing and Data Mining

CSIT — 7th Semester — 2079 — Answer Sheet
Subject Code: CSC410

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Discuss any two drawbacks of Apriori algorithm. Find frequent itemsets and association rules from the transaction database given below using FP-growth algorithm. Assume minimum support is 50% and minimum confidence is 60%.
Transaction ID | Items Purchased
1 | Sausage, Peanut, Beer
2 | Peanut, Beer, Apple
3 | Apple, Milk
4 | Sausage, Peanut, Apple
5 | Sausage, Peanut, Beer, Milk
6 | Sausage, Peanut, Beer, Apple

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. When multilayer perceptron is better choice over other classification algorithms? Consider a multilayer feed-forward neural network given below. Let the learning rate be 0.5. Assume initial values of weights and biases as given in the table below. Train the network for the training tuples (1, 1, 0) and (0, 1, 1), where last number is target output. Show weight and bias updates by using back-propagation algorithm. Assume that activation function is used in the network.
w₁₁ w₁₃ w₂₃ w₂₄ w₃₅ w₄₅ b₃ b₄ b₅
0.2 -0.30 0.50 10.30 6-0.40 8

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Why OLAP operations are used? Discuss various OLAP operation with suitable example of each.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Suppose that we have 5 dimensional data. What will be total number of cuboids generated? If we consider each dimension has 5 levels, what will be the number of cuboids generated?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Discuss different types of attributes with suitable example of each.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Why data normalization is important in data mining? Explain min-max and Z-score normalization approach.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are two categories of hierarchical clustering? Divide the following datapoints into two clusters using agglomerative clustering. {(2,10), (2,5), (8,4), (5,8), (7,5), (6,4)}

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Discuss the concept of K-means and Mini-batch K-means algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is confusion matrix? Discuss various classification measures along with their mathematical formulae.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What are application areas of graph mining? Explain the concept behind inductive logic programming with suitable demonstration.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss the concept of text mining with its practical implications.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write down short notes on a. DataMart b. Market Basket Analysis

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.